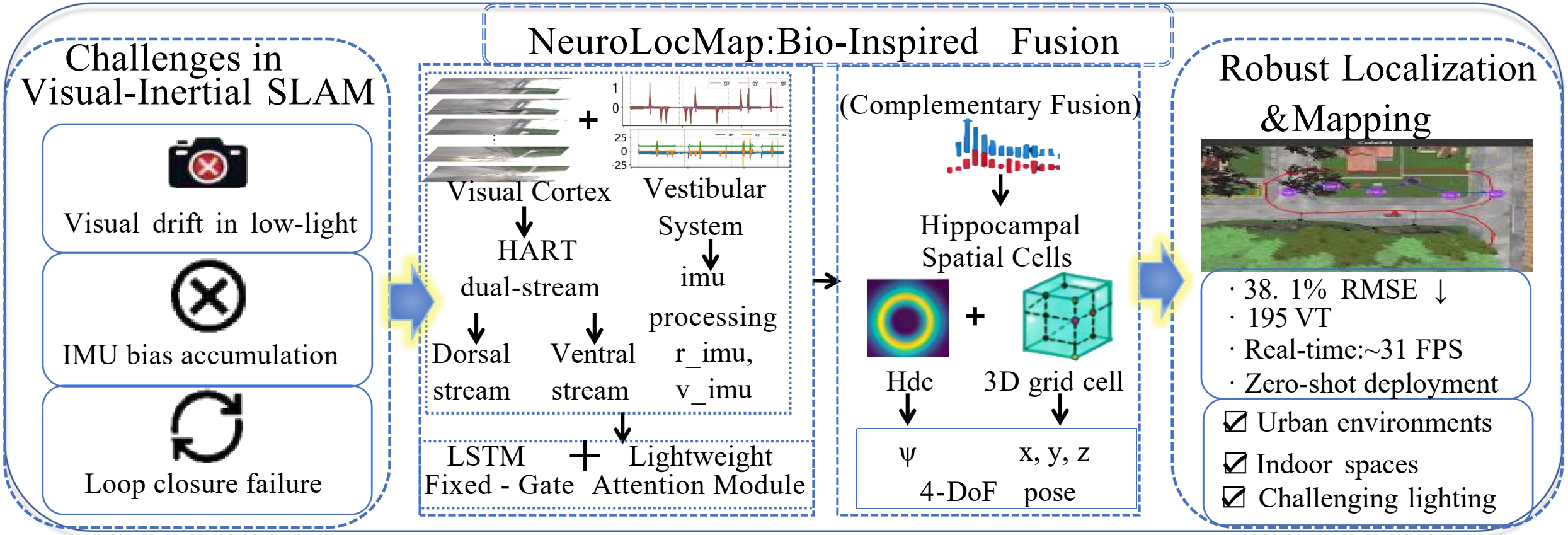
Challenges in Visual-Inertial SLAM
Visual drift in low-light
IMU bias accumulation
Loop closure failure
NeuroLocMap:Bio-Inspired Fusion
Visual Cortex
Vestibular System
HART dual-stream
imu processing
Dorsal stream
Ventral stream
r_imu, v_imu
LSTM Fixed - Gate
Lightweight Attention Module
(Complementary Fusion)
Hippocampal Spatial Cells
Hdc
3D grid cell
ψ
x, y, z
4-DoF pose
Robust Localization &Mapping
· 38. 1% RMSE ↓
· 195 VT
· Real-time:~31 FPS
· Zero-shot deployment
☑ Urban environments
☑ Indoor spaces
☑ Challenging lighting